

GATE DA MASTER GUIDE • 80 QUESTIONS BANK (PYQS & PRACTICE)

Paper: Data Science & Artificial Intelligence (DA)

Standard: 10 PYQs + 10 Practice per Topic (80 Total)

TOPIC 1: PYTHON FUNDAMENTALS

Data Types, Operators & Control Flow • PYQs (1–4)

Q1. GATE DA 2024 • Set Difference & Assignment Order

MCQ • 1 Mark

Consider the execution of the following Python statements:

```
A = {"this", "is", "a", "cat"} B = {"this", "is", "a", "dog"} C = {"that", "was", "a", "horse"} A, B, C = A - B, B - C, C - A print("that" in B)
```

What is the printed output?

- (A) True (B) False
(C) KeyError (D) TypeError

Ans: (B) False

In multiple assignment, right-side expressions evaluate before assignment: $B - C = \{\text{"this", "is", "dog"}\}$. Thus "that" in B is False.

Q2. GATE DA 2024 • Dictionary Key Existence & Membership

MCQ • 1 Mark

What is the output of the following Python code snippet?

```
d = {1: 'a', 2: 'b', 3: 'c'} x = d.get(4, 'd') y = 2 in d z = 'a' in d print(x, y, z)
```

- (A) None True True (B) d True False
(C) d False True (D) KeyError True False

Ans: (B) d True False

$d.get(4, 'd')$ returns default 'd'; $k \text{ in } d$ checks keys only, so $2 \text{ in } d$ is True, but 'a' in d is False.

Q3. GATE DA 2025 • Integer Division & Modulo Arithmetic

NAT • 1 Mark

Consider the following Python expression evaluation:

```
ans = (-17 // 4) + (-17 % 4) + (17 // -4)
```

The numerical integer value of ans is _____.

Ans: -7

Python uses floor division (towards $-\infty$): $-17 // 4 = -5$; $-17 \% 4 = 3$ (since $4 \times (-5) + 3 = -17$); $17 // -4 = -5$. Total = $-5 + 3 + (-5) = -7$.

Q4. GATE DA 2024 • Loop with Break and Else Clause

MCQ • 1 Mark

What will be printed when the following code is executed?

```
s = 0 for i in range(1, 6): if i % 4 == 0: break s += i else: s += 10 print(s)
```

- (A) 6 (B) 16
(C) 15 (D) 25

Ans: (A) 6

Loop iterates for $i=1, 2, 3$ (sum = $1+2+3=6$). When $i=4$, $i \% 4 == 0$ triggers break. The else block is skipped upon break. Prints 6.

Q5. GATE DA 2025 • Tuple Immutability & In-Place Augmented Assignment

MCQ • 1 Mark

Consider the following code:

```
t = (1, 2, [3, 4]) try: t[2] += [5, 6] except Exception as e: print(type(e).__name__, t[2])
```

Which ONE of the following is the printed output?

- (A) TypeError [3, 4, 5, 6] (B) TypeError [3, 4]
(C) ValueError [3, 4] (D) AttributeError [3, 4, 5, 6]

Ans: (A) TypeError [3, 4, 5, 6]

In `t[2] += [5, 6]`, extend mutates the list in-place to `[3, 4, 5, 6]`, but the subsequent assignment to `t[2]` raises `TypeError`.

Q6. GATE DA 2024 • Short-Circuit Boolean Evaluation

MCQ • 1 Mark

What is the exact evaluation result of the following Python expression?

```
ans = (0 and 10) or ("apple" and not "") or (5 and 0) print(ans)
```

- (A) 0 (B) True
(C) "apple" (D) 10

Ans: (B) True

`0 and 10` evaluates to `0` (falsy). In second term: `not ""` is `True`; `"apple" and True` gives `True`. Evaluation short-circuits to `True`.

Q7. GATE DA 2026 Ref • Dictionary Key Hashing Restrictions

MSQ • 2 Marks

Which of the following objects can be used as a **valid key** in a standard Python dictionary?

- (A) `(1, 2, "gate")`
(B) `[1, 2, 3]`
(C) `(1, 2, [3, 4])`
(D) `frozenset([1, 2, 3])`

Ans: (A) and (D)

Dictionary keys must be hashable and immutable. Lists (B) and tuples containing mutable objects like lists (C) are unhashable. Frozensets (D) and tuples of immutables (A) are hashable.

Q8. GATE DA 2024 • String Immutability and Replacement

NAT • 1 Mark

Consider the following code:

```
text = "GATE_DA_2024" text.replace("_", "") ans = len(text) + text.count("_") print(ans)
```

The numerical value printed by the code is _____.

Ans: 14

Python strings are immutable. `text.replace()` returns a new string; `text` remains unchanged. `len("GATE_DA_2024") = 12`, count of `'_'` is 2. Result = `12 + 2 = 14`.

Q9. GATE DA 2025 • Operator Precedence and Bitwise Logic

NAT • 1 Mark

What is the integer value printed by executing the following Python snippet?

```
val = 3 * 2 ** 2 + 5 & 11 print(val)
```

Ans: 1

Precedence: $** > * > + > \&$.

1. $2 ** 2 = 4$; 2. $3 * 4 = 12$; 3. $12 + 5 = 17$; 4. $17 \& 11$ in binary: $10001 \& 01011 = 00001 = 1$.

Q10. GATE DA 2024 • Lambda Function & Map Transformation

MCQ • 1 Mark

Consider the following code:

```
data = [1, 2, 3, 4] res = list(map(lambda x: x**2 if x % 2 == 0 else x * 2, data)) print(res)
```

(A) [2, 4, 6, 16]

(B) [1, 4, 9, 16]

(C) [2, 4, 6, 8]

(D) [4, 4, 12, 16]

Ans: (A) [2, 4, 6, 16]

For 1 (odd): $1*2 = 2$. For 2 (even): $2^2 = 4$. For 3 (odd): $3*2 = 6$. For 4 (even): $4^2 = 16$. Output: [2, 4, 6, 16].

PQ1. Practice • Falsy Coercion & Short-Circuit Chaining

MCQ • 1 Mark

What is the printed output of the following statement?

```
res = ([ ] or { } or [0] or 100) and (None or 42) print(res)
```

(A) 42

(B) [0]

(C) 100

(D) None

Ans: (A) 42

Left side: [] and { } are falsy; [0] is non-empty (truthy), so left side yields [0]. Right side: None is falsy, yielding 42. Since [0] is truthy, [0] and 42 evaluates to 42.

PQ2. Practice • Default Mutable Argument Trap

MCQ • 1 Mark

What is the output of the following Python snippet?

```
def append_item(val, lst=[]): lst.append(val) return lst a = append_item(1) b = append_item(2, [ ]) c = append_item(3) print(a, c)
```

(A) [1] [3]

(B) [1, 3] [1, 3]

(C) [1, 2, 3] [1, 2, 3]

(D) [1] [1, 3]

Ans: (B) [1, 3] [1, 3]

Default list lst is instantiated once at function definition time. Calls a and c share the same default list object in memory, resulting in [1, 3] for both.

NAT • 2 Marks

PQ3. Practice • Nested Control Flow with While-Else

What is the integer value printed by this snippet?

```
n = 12
count = 0
while n > 1:
    if n % 5 == 0:
        count += 5
        break
    n -= 3
    count += 1
else:
    count += 10
print(count)
```

Ans: 14

Trace: $n=12$ ($\text{count}=1, n=9$) $\rightarrow n=9$ ($\text{count}=2, n=6$) $\rightarrow n=6$ ($\text{count}=3, n=3$) $\rightarrow n=3$ ($\text{count}=4, n=0$). Loop terminates because $n > 1$ is false without encountering break. The else branch executes: $4 + 10 = 14$.

NAT • 1 Mark

PQ4. Practice • Set Symmetric Difference Algebra

Let $S1 = \{x \text{ for } x \text{ in range}(10) \text{ if } x \% 2 == 0\}$ and $S2 = \{x \text{ for } x \text{ in range}(10) \text{ if } x \% 3 == 0\}$. What is the length of $S1 \Delta S2$ (symmetric difference)?

Ans: 6

$S1 = \{0, 2, 4, 6, 8\}$ (evens). $S2 = \{0, 3, 6, 9\}$ (multiples of 3). Intersection $S1 \cap S2 = \{0, 6\}$. Symmetric difference $S1 \Delta S2 = \{2, 3, 4, 8, 9, \dots\}$: exactly $(5 + 4 - 2 \times 2) = 5$. Wait: Elements are $\{2, 4, 8\} + \{3, 9\} = 5$ elements. Answer: 5.

MCQ • 1 Mark

PQ5. Practice • Variable Scope and Nonlocal Binding

What is the output of the following function execution?

```
def outer():
    x = 10
    def inner():
        nonlocal x
        x += 5
    return x
return inner() + x
print(outer())
```

(A) 25

(B) 30

(C) 20

(D) UnboundLocalError

Ans: (B) 30

nonlocal `x` rebinds outer's variable `x`. Inside `inner()`, `x` becomes 15 and returns 15. Then `x` in `outer` is also 15. Total = $15 + 15 = 30$.

NAT • 1 Mark

PQ6. Practice • Dictionary Comprehension and Key Collision

Consider the following Python statement:

```
d = {k % 4: k * 2 for k in range(10)}
ans = sum(d.values())
print(ans)
```

The numerical value printed is _____.

Ans: 60

Keys are $k \% 4 \in \{0, 1, 2, 3\}$. For key 0: last $k=8 \rightarrow 16$. For key 1: last $k=9 \rightarrow 18$. For key 2: last $k=6 \rightarrow 12$. For key 3: last $k=7 \rightarrow 14$. Sum of values = $16 + 18 + 12 + 14 = 60$.

PQ7. Practice • Any / All Truth Value Testing

MSQ • 2 Marks

Which of the following expressions evaluate to True in Python?

- (A) `all([1, 2, "False", [0]])`
- (B) `all([])`
- (C) `any([])`
- (D) `any([0, "", {}, ()])`

Ans: (A) and (B)

- (A) All items are non-empty/truthy (including string "False" and list [0]) → True.
- (B) Vacuum truth: `all([])` is defined as True in Python standard library.
- (C) `any([])` is False. (D) All entries in the list are falsy, so `any` returns False.

PQ8. Practice • Generator Expression vs List Comprehension

NAT • 1 Mark

What is the integer output printed by the following code?

```
gen = (x * x for x in range(5)) a = next(gen) b = next(gen) c = sum(gen) print(a + b + c)
```

Ans: 30

Values generated are 0, 1, 4, 9, 16. `next(gen)` yields 0; next yields 1. Remaining values in generator are 4, 9, 16 whose sum is 29. Result = 0 + 1 + 29 = 30.

PQ9. Practice • Walrus Operator & Inline Assignment

NAT • 1 Mark

What is the integer output printed by this snippet?

```
data = [2, 5, 8, 11] s = sum(w := [x + 1 for x in data if x % 2 != 0]) + len(w) print(s)
```

Ans: 20

Odd numbers: 5 and 11. `w = [5+1, 11+1] = [6, 12]`. `sum(w) = 18`, `len(w) = 2`. Result = 18 + 2 = 20.

PQ10. Practice • Chained Comparison Operator Evaluation

MCQ • 1 Mark

What is the output of `print(1 < 2 == 2 > 0 and not 3 == 4)`?

- (A) True
- (B) False
- (C) TypeError
- (D) SyntaxError

Ans: (A) True

Python chains comparisons: `1 < 2 == 2 > 0` is equivalent to `(1 < 2) and (2 == 2) and (2 > 0)`, which is True. `not 3 == 4` is also True. True and True = **True**.

Q11. GATE DA 2024 • In-Place Two-Pointer Array Reversal

MCQ • 1 Mark

Consider the following Python function:

```
def fun(D, s1, s2): if s1 < s2: D[s1], D[s2] = D[s2], D[s1] fun(D, s1 + 1, s2 - 1)
```

What does this function perform on list D?

- (A) Reverses elements between s1 and s2 (B) Sorts elements between s1 and s2
(C) Finds max element in D (D) Circularly shifts elements

Ans: (A) Reverses elements between s1 and s2

Two pointers converge inward from s1 and s2, swapping symmetric elements until they meet or cross.

Q12. GATE DA 2024 • List Modification: Append vs Extend

MCQ • 1 Mark

Let A = [1, 2, 3] and B = [4, 5, 6]. Which ONE of the following statements results in A = [1, 2, 3, 4, 5, 6]?

- (A) A.append(B) (B) A.extend(B)
(C) A.insert(B) (D) A = [A, B]

Ans: (B) A.extend(B)

extend iterates over B appending each element into A in-place.

Q13. GATE DA 2025 • Negative Stride Slicing & Bounds

NAT • 1 Mark

What is the numerical integer value printed by the following code?

```
arr = [10, 20, 30, 40, 50, 60, 70, 80, 90] sub = arr[-2:2:-2] print(sum(sub))
```

Ans: 180

Index -2 is 7 (value 80). Stride is -2, stop is index 2 (exclusive). Indices visited: 7 (80), 5 (60), 3 (40). Sum = 80 + 60 + 40 = **180**.

Q14. GATE DA 2025 • Two-Pointer Convergence (Problem 1: Pair Target Sum)

MCQ • 2 Marks

Given sorted list arr = [2, 4, 7, 11, 15, 19, 23, 28] and target = 22:

```
def pair_search(arr, target): l, r, steps = 0, len(arr) - 1, 0 while l < r: steps += 1 s = arr[l] + arr[r] if s == target: return steps elif s < target: l += 1 else: r -= 1 return steps
```

What is the returned number of steps?

- (A) 3 (B) 4
(C) 5 (D) 6

Ans: (C) 5

1: (2,28)=30>22→r=6. 2: (2,23)=25>22→r=5. 3: (2,19)=21<22→l=1. 4: (4,19)=23>22→r=4. 5: (4,15)=19<22→l=2. Found at next check (7+15=22). Steps = 5.

Q15. GATE DA 2024 • Shallow Copying via Slice Operator

MCQ • 1 Mark

What is the printed output of the following Python snippet?

```
L1 = [1, [2, 3], 4] L2 = L1[:] L2[1][0] = 99 L2[0] = 100 print(L1[0], L1[1][0])
```

(A) 100 99

(B) 1 99

(C) 1 2

(D) 100 2

Ans: (B) 1 99

`L1[:]` creates a shallow copy. Modifying primitive `L2[0]` does not affect `L1[0]` (remains 1), but mutating nested list `L2[1][0]` reflects in `L1`.

Q16. GATE DA 2026 Ref • Two-Pointer (Problem 2: Segregate 0s and 1s)

MCQ • 2 Marks

A list of size N containing binary digits 0 and 1 is segregated using two pointers moving towards each other. What is the tightest asymptotic time and auxiliary space complexity?

(A) $O(N)$ time, $O(1)$ space(B) $O(N \log N)$ time, $O(1)$ space(C) $O(N)$ time, $O(N)$ space(D) $O(N^2)$ time, $O(1)$ space**Ans: (A) $O(N)$ time, $O(1)$ space**

Left pointer skips 0s, right pointer skips 1s, swapping when left points to 1 and right points to 0. Every item is inspected once in $O(N)$ time with $O(1)$ memory.

Q17. GATE DA 2025 Ref • Two-Pointer (Problem 3: Dutch National Flag 3-Way Partition)

NAT • 2 Marks

The function below partitions values into < 2 , $= 2$, and > 2 :

```
def partition3(nums): low, mid, high = 0, 0, len(nums) - 1 swaps = 0 while mid <= high: if
nums[mid] < 2: nums[low], nums[mid] = nums[mid], nums[low] low += 1; mid += 1; swaps += 1 elif nums[mid]
== 2: mid += 1 else: nums[mid], nums[high] = nums[high], nums[mid] high -= 1; swaps += 1 return swaps arr
= [2, 0, 3, 1, 2] print(partition3(arr))
```

The number of swaps executed is _____.

Ans: 31. `mid=0 (2)`: `mid=1`.2. `mid=1 (0)`: `swap(nums[0], nums[1])` → `[0, 2, 3, 1, 2]`, `swaps=1`, `low=1`, `mid=2`.3. `mid=2 (3)`: `swap(nums[2], nums[4])` → `[0, 2, 2, 1, 3]`, `swaps=2`, `high=3`.4. `mid=2 (2)`: `mid=3`.5. `mid=3 (1)`: `swap(nums[1], nums[3])` → `[0, 1, 2, 2, 3]`, `swaps=3`, `low=2`, `mid=4 > high`. Total swaps = **3**.**Q18. GATE DA 2024 • List Pop with Negative Index**

NAT • 1 Mark

What is the integer value printed by this code?

```
nums = [5, 10, 15, 20, 25, 30] a = nums.pop(-2) b = nums.pop(1) print(a + b + len(nums))
```

Ans: 39

`nums.pop(-2)` removes and returns 25. `nums` becomes `[5, 10, 15, 20, 30]`. `nums.pop(1)` removes and returns 10. Remaining length is 4. Result = `25 + 10 + 4` = **39**.

Q19. GATE DA 2025 • Sublist Assignment and Length Shift

NAT • 1 Mark

Consider the following code:

```
A = [1, 2, 3, 4, 5] A[1:4] = [10, 20] print(sum(A))
```

The numerical value printed by the code is _____.

Ans: 36

Slice A[1:4] (elements 2, 3, 4) is replaced by [10, 20]. The list becomes [1, 10, 20, 5]. Sum = 1 + 10 + 20 + 5 = 36.

Q20. GATE DA 2024 • List Indexing Matrix Trace

MCQ • 1 Mark

Given matrix M = [[1, 2], [3, 4], [5, 6]], what is the output of [row[1] for row in M if row[0] > 2]?

(A) [4, 6]

(B) [2, 4]

(C) [3, 5]

(D) [4]

Ans: (A) [4, 6]

Row 1: [1, 2], row[0]=1 > 2. Row 2: [3, 4], row[0]=3 > 2 → selects row[1]=4. Row 3: [5, 6], row[0]=5 > 2 → selects row[1]=6. Yields [4, 6].

PQ11. Practice • Two-Pointer (Problem 1: Remove Duplicates in Sorted Array)

NAT • 2 Marks

The function below executes an in-place two-pointer deduplication:

```
def remove_duplicates(nums): if not nums: return 0 k = 0 for i in range(1, len(nums)): if nums[i] != nums[k]: k += 1 nums[k] = nums[i] return k + 1 nums = [1, 1, 2, 2, 2, 3, 4, 4, 5, 5, 5, 6] ans = remove_duplicates(nums) print(ans + nums[3])
```

Ans: 10

Unique elements are 1, 2, 3, 4, 5, 6 (ans = 6). First 6 positions become [1, 2, 3, 4, 5, 6], so nums[3] = 4. Result = 6 + 4 = 10.

PQ12. Practice • Two-Pointer (Problem 2: Move Zeros to End)

NAT • 2 Marks

Consider the two-pointer code snippet below:

```
def move_zeros(arr): left = 0 swaps = 0 for right in range(len(arr)): if arr[right] != 0: if left != right: arr[left], arr[right] = arr[right], arr[left] swaps += 1 left += 1 return swaps arr = [0, 3, 0, 5, 0, 7, 0, 9] print(move_zeros(arr))
```

Ans: 4

Non-zero values are 3, 5, 7, 9 located at indices 1, 3, 5, 7. In each case left != right, resulting in exactly 4 swaps to indices 0, 1, 2, 3. Output: 4.

PQ13. Practice • Two-Pointer (Problem 3: Container With Most Water)

MCQ • 2 Marks

Given vertical board heights $H = [1, 8, 6, 2, 5, 4, 8, 3, 7]$, a two-pointer algorithm computes $\text{area} = (r - l) * \min(H[l], H[r])$ by advancing the pointer pointing to the shorter line:

- (A) 42 (B) 49
(C) 56 (D) 40

Ans: (B) 49

At $l=1$ ($H=8$) and $r=8$ ($H=7$): width = $8 - 1 = 7$; min height = $\min(8, 7) = 7$. Area = $7 \times 7 = 49$. All other combinations yield area ≤ 40 .

PQ14. Practice • Multi-Dimensional Slicing and Negative Index

NAT • 1 Mark

What is the integer value printed by this code?

```
matrix = [ [1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12] ]
diag = [matrix[i][-i + 1] for i in range(3)]
print(sum(diag))
```

Ans: 20

- $i=0$: $\text{matrix}[0][-1] = 4$
 - $i=1$: $\text{matrix}[1][-2] = 7$
 - $i=2$: $\text{matrix}[2][-3] = 9$
- Sum = $4 + 7 + 9 = 20$.

PQ15. Practice • List In-Place Reverse vs Built-in Reversed

MCQ • 1 Mark

Consider the following Python snippet:

```
L = [10, 20, 30]
x = L.reverse()
y = list(reversed(L))
print(x, y)
```

- (A) None [10, 20, 30] (B) [30, 20, 10] [10, 20, 30]
(C) None [30, 20, 10] (D) [30, 20, 10] None

Ans: (A) None [10, 20, 30]

`L.reverse()` reverses list in-place and returns None (so $x = \text{None}$, $L = [30, 20, 10]$). `reversed(L)` then reverses it back, yielding $[10, 20, 30]$.

PQ16. Practice • Two-Pointer Sorted Array Intersection

NAT • 2 Marks

Given two sorted lists $A = [1, 2, 4, 5, 6]$ and $B = [2, 3, 5, 7]$, a two-pointer approach finds their intersection. What is the sum of the intersected elements?

Ans: 7

Two pointers advance comparing $A[i]$ and $B[j]$. Common elements are 2 and 5. Sum = $2 + 5 = 7$.

PQ17. Practice • Two-Pointer Trapping Rainwater Core Logic

NAT • 2 Marks

Consider elevations [3, 0, 2, 0, 4]. Using two pointers tracking left_max and right_max, how many total units of water are trapped between bars?

Ans: 7

- Index 1 (h=0): $\min(3, 4) - 0 = 3$ units.
 - Index 2 (h=2): $\min(3, 4) - 2 = 1$ unit.
 - Index 3 (h=0): $\min(3, 4) - 0 = 3$ units.
- Total trapped water = $3 + 1 + 3 = 7$.

PQ18. Practice • List Sorting Key with Lambda Function

MCQ • 1 Mark

What is the printed result of the following Python code?

```
data = ["apple", "pie", "banana", "kiwi"] data.sort(key=lambda s: (len(s), s)) print(data[1])
```

- (A) kiwi (B) pie
(C) apple (D) banana

Ans: (A) kiwi

Sorted by length first, then alphabetically: len 3 ("pie"), len 4 ("kiwi"), len 5 ("apple"), len 6 ("banana"). Element at index 1 is **kiwi**.

PQ19. Practice • Array Flattening via List Comprehension

NAT • 1 Mark

What is the output of `len([x for sub in [[1, 2], [3], [4, 5, 6]] for x in sub if x % 2 != 0])`?

Ans: 3

Flattened elements: 1, 2, 3, 4, 5, 6. Odd elements: 1, 3, 5. Length is **3**.

PQ20. Practice • List Multiplication Mutability Caveat

NAT • 1 Mark

Consider the following code:

```
grid = [[0] * 2] * 3 grid[0][0] = 5 print(sum([row[0] for row in grid]))
```

The numerical value printed by the code is _____.

Ans: 15

`[[0]*2]*3` creates 3 references pointing to the exact same inner list. Mutating `grid[0][0] = 5` mutates all 3 rows. Sum = $5 + 5 + 5 = 15$.

Q21. GATE DA 2024 • Automated Taxi Driver PEAS Specification

MCQ • 1 Mark

Match the PEAS components for an *Automated Taxi Driver*:

P. Performance Measure	1. Steering, Accelerator, Brake, Display
E. Environment	2. Roads, Traffic, Pedestrians, Weather
A. Actuators	3. Safety, Speed, Legality, Comfort, Profits
S. Sensors	4. Cameras, LIDAR, GPS, Speedometer
(A) P-3, E-2, A-1, S-4	(B) P-2, E-3, A-4, S-1
(C) P-3, E-1, A-2, S-4	(D) P-4, E-2, A-1, S-3

Ans: (A) P-3, E-2, A-1, S-4

P represents goals/metrics (3); E is physical operational domain (2); A is physical control mechanisms (1); S is input hardware (4).

Q22. GATE DA 2025 • Intelligent Agent Architecture Taxonomy

MSQ • 2 Marks

Which of the following statements regarding **Agent Architectures** is/are **TRUE**?

- (A) A simple reflex agent selects actions based solely on current percept, ignoring history.
- (B) Goal-based agents combine goal descriptions with state models to plan action sequences.
- (C) Utility-based agents use an explicit utility function to trade off conflicting objectives.
- (D) Model-based reflex agents do not maintain any internal representation of unobserved state.

Ans: (A), (B), and (C)

(A), (B), and (C) are fundamental definitions. (D) is false because model-based agents explicitly maintain internal state models to track unperceived aspects.

Q23. GATE DA 2024 • Environment Properties: Chess with a Clock

MCQ • 1 Mark

How is the task environment of **Chess with a clock** classified?

- (A) Fully observable, Strategic, Semi-dynamic, Discrete, Multi-agent
- (B) Partially observable, Stochastic, Dynamic, Continuous, Single-agent
- (C) Fully observable, Stochastic, Static, Discrete, Multi-agent
- (D) Partially observable, Deterministic, Static, Continuous, Multi-agent

Ans: (A) Fully observable, Strategic, Semi-dynamic, Discrete, Multi-agent

Board is completely visible (fully observable); clock ticks down during deliberation (semi-dynamic); discrete moves; competitive 2-player (multi-agent).

Q24. GATE DA 2026 Ref • Rationality vs Omniscience Principle

MCQ • 1 Mark

In artificial intelligence, what distinguishes an **omniscient agent** from a **rational agent**?

- (A) Omniscient agent knows actual outcome; rational agent maximizes expected performance
- (B) Rational agent has unlimited computation; omniscient agent does not
- (C) Rational agent never makes mistakes; omniscient agent may make errors
- (D) Omniscience is achievable in partially observable environments

Ans: (A) Omniscient agent knows actual outcome; rational agent maximizes expected performance

Rationality maximizes expected performance given percept history and built-in knowledge; omniscience requires knowing actual future outcomes in advance.

Q25. GATE DA 2025 • Learning Agent Structural Decomposition

MCQ • 1 Mark

In a learning agent architecture, which component is responsible for suggesting actions that lead to new experiences and exploration?

- (A) Critic (B) Learning Element
(C) Performance Element (D) Problem Generator

Ans: (D) Problem Generator

The Problem Generator suggests exploratory actions (suboptimal in the short term) to discover better actions in the long run.

Q26. GATE DA 2024 • Medical Diagnosis PEAS Specification

MSQ • 2 Marks

For an AI-based **Medical Diagnosis System**, which of the following mappings is/are **CORRECT** under PEAS?

- (A) **Performance Measure:** Healthy patient, minimize costs, minimize malpractice suits.
(B) **Environment:** Patient, hospital staff, medical testing laboratories.
(C) **Actuators:** Keyboard, keyboard touchscreen, optical barcode scanner.
(D) **Sensors:** Symptoms reported, laboratory test measurements, physician findings.

Ans: (A), (B), and (D)

(C) is incorrect because keyboard and barcode scanners are input devices (Sensors), whereas Actuators are screen displays, prescription printers, and alert signals.

Q27. GATE DA 2026 Ref • Vacuum Cleaner World Agent Function

NAT • 1 Mark

Consider a vacuum-cleaner world with 2 locations (A and B), each of which can be Dirty or Clean. The agent can perceive its location and status (4 possible percepts). If the agent has 3 possible actions (Left, Right, Suck), how many distinct **pure simple reflex agent functions** are mathematically possible?

Ans: 81

There are 4 distinct states/percepts. For each percept, the agent can map to any of the 3 actions independently. Total functions = $3^4 = 81$.

Q28. GATE DA 2024 • Episodic vs Sequential Environments

MCQ • 1 Mark

Which of the following task environments is **episodic**?

- (A) Defective part detection on an assembly line (B) Playing a game of Go
(C) Navigating an autonomous vehicle through city traffic (D) Conversational chatbot negotiating a contract

Ans: (A) Defective part detection on an assembly line

In part classification, each decision depends only on the current image and does not affect the next item arriving on the conveyor belt.

Q29. GATE DA 2025 • Exploration vs Exploitation in Problem Generator

MCQ • 1 Mark

Why does the **Problem Generator** component of a learning agent intentionally suggest suboptimal actions?

- (A) To explore new states and discover globally superior strategies
- (B) Because the performance element has failed
- (C) To simplify the critic's feedback computation
- (D) To reduce memory overhead in the internal model

Ans: (A) To explore new states and discover globally superior strategies

Without exploration, an agent gets trapped in local optima by perpetually exploiting familiar actions.

Q30. GATE DA 2024 • Deterministic vs Stochastic Environments

MCQ • 1 Mark

An environment is termed **deterministic** if:

- (A) The next state is completely determined by current state and agent's action
- (B) The agent has complete sensors covering the whole world
- (C) The environment does not change while the agent deliberates
- (D) There is only one agent operating in the environment

Ans: (A) The next state is completely determined by current state and agent's action

If uncertainty arises solely from incomplete observation, the environment is deterministic but partially observable.

PQ21. Practice • PEAS for Automated Warehouse Robot

MCQ • 1 Mark

Under PEAS, which of the following constitutes an **Actuator** for an automated parcel-sorting robotic arm?

- (A) Pneumatic gripper and joint electric motors
- (B) High-resolution optical barcode scanner
- (C) Percentage of packages sorted correctly per hour
- (D) Conveyor belts and human packaging workers

Ans: (A) Pneumatic gripper and joint electric motors

Grippers and motors perform the physical actions (Actuators). Scanner is Sensor; sorting rate is Performance; conveyor/workers are Environment.

PQ22. Practice • Agent Environment Matrix Classification

MSQ • 2 Marks

Which of the following environments is/are **Partially Observable**?

- (A) Poker (card game with hidden opponent hands)
- (B) Autonomous driving on public highways
- (C) Standard 8-puzzle sliding tile board
- (D) Tic-Tac-Toe

Ans: (A) and (B)

Opponents' cards in Poker and blind spots/pedestrian intents in driving are unobservable. 8-puzzle and Tic-Tac-Toe states are fully visible.

PQ23. Practice • Utility-Based vs Goal-Based Trade-offs

MCQ • 1 Mark

Why is a utility-based agent preferred over a goal-based agent in autonomous navigation?

- (A) Utility functions allow quantitative trade-offs between conflicting goals (e.g., speed vs safety) (B) Goal-based agents cannot find paths to destinations
(C) Utility agents do not require an internal world model (D) Utility agents execute in $O(1)$ time without search

Ans: (A) Utility functions allow quantitative trade-offs between conflicting goals

A goal only provides binary satisfaction (reached or not), whereas utility assigns real numbers to capture degree of desirability and risk.

PQ24. Practice • Continuous vs Discrete Environment

MCQ • 1 Mark

Which of the following environments has a **continuous state space and continuous action space**?

- (A) Autonomous drone attitude and trajectory flight control (B) Sudoku puzzle solver
(C) Chess game playing engine (D) Vacuum cleaner with 4 grid rooms

Ans: (A) Autonomous drone attitude and trajectory flight control

Position, velocity, steering angles, and thrust are continuous real-valued variables in time and space.

PQ25. Practice • Critic Component in Learning Agents

MCQ • 1 Mark

In Russell & Norvig's learning agent architecture, the **Critic** evaluates the agent's behavior against:

- (A) An external, fixed performance standard (B) The problem generator's suggestions
(C) The learning element's internal weights (D) The size of the percept history table

Ans: (A) An external, fixed performance standard

The critic observes sensory percepts and evaluates them against an external standard to generate a reward or penalty signal.

PQ26. Practice • Dynamic vs Semidynamic vs Static

MSQ • 2 Marks

Which of the following statements is/are **CORRECT** regarding environmental temporal dynamism?

- (A) A static environment does not change while the agent is deliberating.
(B) If the environment itself does not change with time, but the agent's performance score does, it is semidynamic.
(C) Chess without a clock is static.
(D) Driving a car is static because the road geometry is fixed.

Ans: (A), (B), and (C)

(D) is false because traffic, pedestrians, and moving obstacles change continuously while the vehicle deliberates (dynamic).

PQ27. Practice • Simple Reflex Infinite Loop Vulnerability

MCQ • 1 Mark

Why can a randomized simple reflex agent outperform a deterministic simple reflex agent in partially observable environments?

- (A) Randomization helps break infinite loops caused by identical percepts
 (B) Randomization expands the agent's sensor range
 (C) Randomization converts partial observability into full observability
 (D) Randomized agents do not require condition-action rules

Ans: (A) Randomization helps break infinite loops caused by identical percepts

In partially observable environments, two different states may yield identical percepts; a deterministic reflex agent can get stuck looping indefinitely.

PQ28. Practice • Competitive vs Cooperative Multi-Agent Classification

MCQ • 1 Mark

Which of the following environments represents a **purely competitive multi-agent** environment?

- (A) Two-player zero-sum Chess
 (B) Coordinated multi-robot search and rescue
 (C) Automated freeway driving with connected vehicles
 (D) Assembly factory robotic team

Ans: (A) Two-player zero-sum Chess

One player's gain is exactly equal to the other's loss (zero-sum).

PQ29. Practice • Model-Based State Update Mechanism

MCQ • 1 Mark

To maintain an internal state, a **model-based reflex agent** requires knowledge about:

- (A) How the world evolves independently, and how the agent's actions affect the world
 (B) The complete historical utility function scores
 (C) A table mapping all conceivable percept sequences directly to actions
 (D) The adversary's neural network weights

Ans: (A) How the world evolves independently, and how actions affect the world

Transition model: $P(s' | s, a)$ and sensor model: $P(e | s')$.

PQ30. Practice • PEAS for Interactive English Tutoring Agent

NAT • 1 Mark

An interactive English tutor agent scores student essays across 4 dimensions (Grammar, Vocabulary, Coherence, Relevance), each graded from 0 to 25. How many distinct performance measure values can the overall composite sum take?

Ans: 101

The sum ranges from minimum $0 \times 4 = 0$ to maximum $25 \times 4 = 100$. Total integer values = $100 - 0 + 1 = 101$.

NAT • 2 Marks

Q31. GATE DA 2024 • 10-Line Python Tracing: Collatz Sequence Step Count

Consider the following 10-line Python snippet:

```
def computeS(n): c = 0 while n > 1: if n % 2 == 0: n = n // 2 else: n = 3 * n + 1 c += 1 return c
print(computeS(7))
```

The integer value printed by the code is _____.

Ans: 16Trajectory: $7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. Exactly **16** steps taken.

MCQ • 2 Marks

Q32. GATE DA 2024 • 10-Line Python Tracing: Dictionary Tree Traversal

What is the integer output printed by this 10-line script?

```
def count(d, i): if i not in d: return 1 ans = 1 for j in d[i]: ans += count(d, j) return ans
tree = {0: [1, 2], 1: [3, 4, 5], 2: [6, 7, 8]} print(count(tree, 0))
```

- (A) 6 (B) 8
(C) 9 (D) 10

Ans: (C) 9Leaves 3, 4, 5, 6, 7, 8 return 1 each. Node 1 returns $1+3=4$. Node 2 returns $1+3=4$. Node 0 returns $1+4+4=9$.

NAT • 2 Marks

Q33. GATE DA 2025 • 10-Line Python Tracing: Matrix In-Place Transposition Check

Consider this 10-line script:

```
M = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] total = 0 for i in range(3): for j in range(i, 3): M[i][j],
M[j][i] = M[j][i], M[i][j] for r in range(3): total += M[r][0] * (10 ** (2 - r)) print(total)
```

The numerical value printed is _____.

Ans: 123Matrix is transposed in-place. First column becomes $[1, 2, 3]$. Total = $1 \times 100 + 2 \times 10 + 3 \times 1 = 123$.

NAT • 2 Marks

Q34. GATE DA 2024 • 10-Line Python Tracing: Nested List Mutation

What is the integer output printed by this 10-line Python snippet?

```
def update(lst): for idx, item in enumerate(lst): if item % 2 == 0: lst[idx] = item // 2 else:
lst[idx] = item * 3 + 1 nums = [2, 7, 4, 3] update(nums) print(sum(nums))
```

Ans: 35

- idx 0 (2): $2//2 = 1$
- idx 1 (7): $7 \times 3 + 1 = 22$
- idx 2 (4): $4//2 = 2$
- idx 3 (3): $3 \times 3 + 1 = 10$

Sum = $1 + 22 + 2 + 10 = 35$.

NAT • 2 Marks

Q35. GATE DA 2025 • 10-Line Python Tracing: Recursive GCD Computation

What is the final integer output printed by this 10-line program?

```
def mystery(a, b, step): if b == 0: return a, step return mystery(b, a % b, step + 1) val, steps =
mystery(105, 252, 0) print(val * 10 + steps)
```

Ans: 214Trace: (105, 252, 0) → (252, 105, 1) → (105, 42, 2) → (42, 21, 3) → (21, 0, 4). Returns (21, 4). Result = 21×10 + 4 = **214**.

NAT • 2 Marks

Q36. GATE DA 2026 Ref • 10-Line Python Tracing: Nested Loop Filtering

What is the numerical output of this 10-line Python snippet?

```
res = 0 for i in range(1, 4): for j in range(1, 4): if (i * j) % 2 != 0: res += (i + j) else: res
-= 1 print(res)
```

Ans: 10Total pairs = 9. $i*j$ is odd only when both i and j are odd: (1,1)→+2, (1,3)→+4, (3,1)→+4, (3,3)→+6. Sum = +16. Remaining 5 pairs subtract 1 each (-5). Result = 16 - 5 = **11**. Wait: 1+1=2, 1+3=4, 3+1=4, 3+3=6 → sum = 16. Even pairs: (1,2), (2,1), (2,2), (2,3), (3,2) = 5 pairs → 16 - 5 = **11**.

MCQ • 1 Mark

Q37. GATE DA 2024 • 10-Line Python Tracing: String Character Frequency Accumulator

Consider this 10-line code snippet:

```
def build_map(s): counts = {} for ch in s: counts[ch] = counts.get(ch, 0) + 1 return "".join(f"{k}
{v}" for k, v in sorted(counts.items())) res = build_map("datascience") print(res[:4])
```

(A) a2c2

(B) a2c1

(C) d1a2

(D) c2a2

Ans: (A) a2c2Sorted characters: 'a' count=2, 'c' count=2, 'd' count=1, 'e' count=2, etc. First four characters formatted are **a2c2**.

NAT • 2 Marks

Q38. GATE DA 2025 • 10-Line Python Tracing: Dynamic Programming Memoization

What is the integer value printed by this 10-line Python snippet?

```
def fib(n, memo={}): if n in memo: return memo[n] if n <= 2: return 1 memo[n] = fib(n - 1, memo) +
fib(n - 2, memo) return memo[n] ans1 = fib(5) ans2 = len(fib.__defaults__[0]) print(ans1 + ans2)
```

Ans: 8fib(5) = 5. Stored keys in default memo dictionary are 3, 4, 5 (length = 3). Result = 5 + 3 = **8**.

Q39. GATE DA 2024 • 10-Line Python Tracing: Bitwise Right Shift Loop

NAT • 1 Mark

What is the output printed by this 10-line program?

```
num = 53 count = 0 while num > 0: if num & 1: count += 1 num = num >> 1 print(count)
```

Ans: 4

Binary representation of 53 is 110101_2 , which contains exactly four 1-bits (hamming weight = 4). Prints 4.

Q40. GATE DA 2026 Ref • 10-Line Python Tracing: Cumulative Prefix Maxima

NAT • 2 Marks

Consider this 10-line Python snippet:

```
arr = [3, 1, 4, 1, 5, 9, 2, 6] leaders = [] curr_max = -1 for x in arr: if x > curr_max:
leaders.append(x) curr_max = x print(sum(leaders))
```

The numerical value printed is _____.

Ans: 21

Elements that establish a strictly new maximum are: 3, 4, 5, 9. Sum = $3 + 4 + 5 + 9 = 21$.

PQ31. Practice • 10-Line Python Tracing: 2D Matrix Grid Transformations

NAT • 2 Marks

What is the final integer value printed by this 10-line Python snippet?

```
grid = [[i * 2 + j for j in range(3)] for i in range(3)] total = 0 for r in range(3): for c in
range(3): if (r + c) % 2 == 0: grid[r][c] *= 2 if r == c: total += grid[r][c] print(total)
```

Ans: 18

On diagonal $r=c$, $r+c$ is always even, so diagonal entries double: (0,0): $0 \times 2=0$; (1,1): $3 \times 2=6$; (2,2): $6 \times 2=12$. Total = $0 + 6 + 12 = 18$.

PQ32. Practice • 10-Line Python Tracing: Recursive Alternating Sign Accumulator

NAT • 2 Marks

What is the integer value printed by this 10-line recursive function?

```
def solve(arr, n, k): if n == 0: return 0 if arr[n - 1] % k == 0: return arr[n - 1] + solve(arr, n
- 1, k) else: return solve(arr, n - 1, k) - arr[n - 1] data = [3, 4, 6, 8, 9, 12] print(solve(data,
len(data), 3))
```

Ans: 18

Multiples of 3 (added): 3, 6, 9, 12 (sum = 30). Non-multiples (subtracted): 4, 8 (sum = 12). Result = $30 - 12 = 18$.

NAT • 2 Marks

PQ33. Practice • 10-Line Python Tracing: Two-Way Queue Simulation

What is the printed integer value of total?

```
Q = [10, 20, 30, 40, 50] total = 0
for k in range(3):
    if k % 2 == 0:
        val = Q.pop(0)
    else:
        val = Q.pop(-1)
    total += val
print(total + len(Q))
```

Ans: 82

• k=0: pops Q[0]=10 → Q=[20, 30, 40, 50]
 • k=1: pops Q[-1]=50 → Q=[20, 30, 40]
 • k=2: pops Q[0]=20 → Q=[30, 40]
 total = 10 + 50 + 20 = 80. Remaining length is 2. Result = 80 + 2 = **82**.

NAT • 2 Marks

PQ34. Practice • 10-Line Python Tracing: String Run-Length Encoder

What is the integer value printed by this 10-line Python snippet?

```
s = "AAABBCDDD" res = [] count = 1
for i in range(1, len(s)):
    if s[i] == s[i - 1]:
        count += 1
    else:
        res.append(count)
        count = 1
res.append(count)
print(sum(res) * len(res))
```

Ans: 36

Runs: A(3), B(2), C(1), D(3). res = [3, 2, 1, 3]. sum(res) = 9 (total characters). len(res) = 4 (distinct runs). Result = 9 × 4 = **36**.

NAT • 2 Marks

PQ35. Practice • 10-Line Python Tracing: Fast Exponentiation Steps

Consider this 10-line power computation function:

```
def power(x, n):
    res, mults = 1, 0
    while n > 0:
        if n % 2 == 1:
            res *= x
        mults += 1
        x *= x
        n = n // 2
    return res
print(power(2, 13))
```

The number of multiplications executed is _____.

Ans: 7

13 in binary is 1101_2 (4 bits, three 1s). The loop executes 4 times: $x *= x$ executes 4 times; $res *= x$ executes 3 times. Total multiplications = 4 + 3 = **7**.

NAT • 2 Marks

PQ36. Practice • 10-Line Python Tracing: Nested Dictionary Inversion

What is the integer output printed by this 10-line Python snippet?

```
data = {'p': 3, 'q': 1, 'r': 3, 's': 2} inv = {}
for k, v in data.items():
    if v not in inv:
        inv[v] = []
    inv[v].append(k)
total = sum(k * len(v) for k, v in inv.items())
print(total)
```

Ans: 9

inv = {3: ['p', 'r'], 1: ['q'], 2: ['s']}. Total = 3×2 + 1×1 + 2×1 = 6 + 1 + 2 = **9**.

NAT • 2 Marks

PQ37. Practice • 10-Line Python Tracing: Pascal Triangle Row Accumulator

What is the integer value printed by this 10-line script?

```
def get_row(n): row = [1]
for _ in range(n): row = [x + y for x, y in zip([0] + row, row + [0])]
return row
ans = sum(get_row(5))
print(ans)
```

Ans: 32

The sum of elements in the n-th row of Pascal's triangle is 2^n . For n=5: $2^5 = 32$.

NAT • 2 Marks

PQ38. Practice • 10-Line Python Tracing: Custom Filter & Reduce

What is the numerical output printed by this 10-line program?

```
arr = [1, 2, 3, 4, 5, 6]
acc = 1
for x in arr:
    if x % 2 == 0:
        acc = acc * x - 1
    else:
        acc += x
print(acc)
```

Ans: 107

Start acc=1.

- x=1 (odd): $1+1 = 2$
- x=2 (even): $2 \times 2 - 1 = 3$
- x=3 (odd): $3+3 = 6$
- x=4 (even): $6 \times 4 - 1 = 23$
- x=5 (odd): $23+5 = 28$
- x=6 (even): $28 \times 6 - 1 = 168 - 1 = 167$.

NAT • 2 Marks

PQ39. Practice • 10-Line Python Tracing: Cycle Detection in Array Mapping

Consider this 10-line indexing cycle trace:

```
P = [2, 0, 1, 4, 3]
visited = set()
curr, steps = 0, 0
while curr not in visited:
    visited.add(curr)
    curr = P[curr]
    steps += 1
print(steps * 10 + curr)
```

Ans: 30

Path: $0 \rightarrow P[0]=2 \rightarrow P[2]=1 \rightarrow P[1]=0$ (cycle closes at node 0). Steps = 3. Visited node repeated is 0. Result = $3 \times 10 + 0 = 30$.

NAT • 2 Marks

PQ40. Practice • 10-Line Python Tracing: String Stack Parentheses Counter

What is the integer value printed by this 10-line Python snippet?

```
expr = "(()())((()))"
depth, max_depth = 0, 0
for ch in expr:
    if ch == '(':
        depth += 1
    if depth > max_depth:
        max_depth = depth
    else:
        depth -= 1
print(max_depth)
```

Ans: 3

Maximum nesting of parentheses occurs during the sub-expression `((()))`, reaching depth 3.

MASTER ANSWER KEY • COMPLETE 80 QUESTIONS SUMMARY

GATE DA Question Bank: Python Fundamentals, Two-Pointer, PEAS & Tracing

Standardized Solutions

PART I: PREVIOUS YEAR QUESTIONS (PYQs 1 to 40)											
Q#	Type	Key	Q#	Type	Key	Q#	Type	Key	Q#		
Q1	MCQ	(B) False	Q11	MCQ	(A) Reverse	Q21	MCQ	(A) P-3,E-2,A-1,S-4	Q31	NAT	16
Q2	MCQ	(B) d True False	Q12	MCQ	(B) extend	Q22	MSQ	A, B, C	Q32	MCQ	(C) 9
Q3	NAT	-7	Q13	NAT	180	Q23	MCQ	(A) Semi-dynamic	Q33	NAT	123
Q4	MCQ	(A) 6	Q14	MCQ	(C) 5	Q24	MCQ	(A) Rationality	Q34	NAT	35
Q5	MCQ	(A) TypeError	Q15	MCQ	(B) 1 99	Q25	MCQ	(D) Problem Gen	Q35	NAT	214
Q6	MCQ	(B) True	Q16	MCQ	(A) O(N), O(1)	Q26	MSQ	A, B, D	Q36	NAT	11
Q7	MSQ	A, D	Q17	NAT	3	Q27	NAT	81	Q37	MCQ	(A) a2c2
Q8	NAT	14	Q18	NAT	39	Q28	MCQ	(A) Assembly Part	Q38	NAT	8
Q9	NAT	1	Q19	NAT	36	Q29	MCQ	(A) Explore Optima	Q39	NAT	4
Q10	MCQ	(A) [2, 4, 6, 16]	Q20	MCQ	(A) [4, 6]	Q30	MCQ	(A) Deterministic	Q40	NAT	21

PART II: PRACTICE QUESTIONS (PQs 1 to 40)											
PQ#	Type	Key	PQ#	Type	Key	PQ#	Type	Key	PQ#		
PQ1	MCQ	(A) 42	PQ11	NAT	10	PQ21	MCQ	(A) Gripper/Motors	PQ31	NAT	18
PQ2	MCQ	(B) [1, 3]	PQ12	NAT	4	PQ22	MSQ	A, B	PQ32	NAT	18
PQ3	NAT	14	PQ13	MCQ	(B) 49	PQ23	MCQ	(A) Trade-offs	PQ33	NAT	82
PQ4	NAT	5	PQ14	NAT	20	PQ24	MCQ	(A) Drone Flight	PQ34	NAT	36
PQ5	MCQ	(B) 30	PQ15	MCQ	(A) None [10,20,30]	PQ25	MCQ	(A) External Std	PQ35	NAT	7
PQ6	NAT	60	PQ16	NAT	7	PQ26	MSQ	A, B, C	PQ36	NAT	9
PQ7	MSQ	A, B	PQ17	NAT	7	PQ27	MCQ	(A) Break Loops	PQ37	NAT	32
PQ8	NAT	30	PQ18	MCQ	(A) kiwi	PQ28	MCQ	(A) 2-Player Chess	PQ38	NAT	167
PQ9	NAT	20	PQ19	NAT	3	PQ29	MCQ	(A) World Transition	PQ39	NAT	30
PQ10	MCQ	(A) True	PQ20	NAT	15	PQ30	NAT	101	PQ40	NAT	3

Exam Notes: Numerical Answer Type (NAT) questions require exact numeric entry without rounding unless specified. Multiple Select Questions (MSQ) have one or more correct options with no negative marking. All questions adhere directly to the official GATE DA syllabus standards.